

Maximal Three-Independent Subsets of $\{0, 1, 2\}^n$

A.R. CALDERBANK AND P.C. FISHBURN

Mathematical Sciences Research Center, AT&T Bell Laboratories, Murray Hill, NJ 07974

Communicated by S. Vanstone

Received February 18, 1993.

Abstract. We consider a variant of the classical problem of finding the size of the largest cap in the r -dimensional projective geometry $\text{PG}(r, 3)$ over the field IF_3 with 3 elements. We study the maximum size $f(n)$ of a subset S of IF_3^n with the property that the only solution to the equation $x_1 + x_2 + x_3 = \mathbf{0}$ is $x_1 = x_2 = x_3$. Let $c_n = f(n)^{1/n}$ and $c = \sup\{c_1, c_2, \dots\}$. We prove that $c > 2.21$, improving the previous lower bound of 2.1955....

1. Introduction

This paper considers a variant of the classical packing problem for the r -dimensional projective geometry $\text{PG}(r, q)$ over the field IF_q with q elements. This classical problem is to find the maximum cardinality of a set of points with the property that no d points from this set are linearly dependent. When $d = 3$, the sets in question are called *caps*. The packing problem was first formalized by Bose [1], and subsequently Segré [2], [3] obtained comprehensive upper and lower bounds.

The underlying vector space provides a more natural framework for the problem we consider here than does the projective geometry itself. We ask for the maximum size $f(n)$ of a subset S of IF_3^n with the *independence* property that the only solution to the equation $x_1 + x_2 + x_3 = \mathbf{0}$ is $x_1 = x_2 = x_3$. It is easily seen that the limiting behavior of the size of a maximal cap in $\text{PG}(r, 3)$ is determined by the limiting behavior of $f(n)$.

Let $c_n = f(n)^{1/n}$ and let $c = \sup\{c_1, c_2, \dots\}$. The main open question about c is whether $c < 3$. We do not resolve this, but we do derive a lower bound that exceeds the largest previously known lower bound for c of $112^{1/6} = 2.1955$. A construction described in Section 3 shows that $c > 2.21$. We precede this in Section 2 with observations about $f(n)$ and extremal independent sets for small n . Larger lower bounds than known previously for $f(5)$ and $f(7)$ are verified, and our construction for $n = 7$ is exploited in Section 3 to obtain the new bound on c .

Central to the new construction is the 56 point cap in $\text{PG}(5, 3)$ that has been studied by Segré [4], McLaughlin [5], Hill [6], [7], [8], Bruen and Hirschfeld [9] and Games [10]. The following description is taken from Calderbank and Kantor [11].

Let $e_\infty, e_0, e_1, e_2, e_3, e_4$ be a basis for a 6-dimensional vector space over $\text{GF}(3)$. All vectors will be written in terms of their coordinates with respect to this basis. Our $\mathbb{Z}_2^5$ consists of all diagonal transformations, $\text{diag}(\pm 1, \pm 1, \pm 1, \pm 1, \pm 1, \pm 1)$, with determinant 1. The group $\text{PSL}_2(5)$ represented by all transformations

$$x \rightarrow \frac{ax + b}{cx + d}, \quad ad - bc = 1 \quad (a, b, c, d \in \text{GF}(5)),$$

acts on $\{e_\infty, e_0, e_1, e_2, e_3, e_4\}$ in the obvious way while normalizing our $\mathbb{Z}_2^5$. The resulting group $\mathbb{Z}_2^5 \rtimes \text{PSL}_2(5)$ sends $\langle(111000)\rangle$ to 40 points and $\langle(111111)\rangle$ to 16 points. The 56 point cap is the union of these two orbits. Hill [6] showed uniqueness of this cap. It follows from the existence of this cap that $c \geq c_6 \geq 112^{1/6} = 2.1955\dots$

Frankl, Graham and Rödl [12] were unaware of the connections with finite geometry and obtained the bound $c \geq 2.179\dots$ by different methods.

At this point we introduce terminology and notation. Let $\text{IF}_p^n = \{0, 1, \dots, p-1\}^n$ denote the usual additive (sums mod p) group of order p^n with identity $\mathbf{0} = (0, \dots, 0)$ and inverse x^- of x with $x + x^- = \mathbf{0}$ for all $x \in \text{IF}_p^n$. A subset S of IF_p^n is *k-independent* if for all n -tuples $x_1, x_2, \dots, x_k$ in S ,

$$\sum_{i=1}^k x_i = \mathbf{0} \Rightarrow x_1 = x_2 = \dots = x_k.$$

We are interested in the largest k -independent subsets of IF_p^n for $(k, p) \geq (2, 2)$ and $n = 1, 2, \dots$. Because definitive results are known for $k = 2$ and $(k, p) = (3, 2)$, we focus on $(k, p) = (3, 3)$, where surprisingly little is known about maximum or *extremal* independent sets.

When p is even, $x = x^- \Leftrightarrow x \in \{\mathbf{0}, (p/2, \dots, p/2)\}$; when p is odd, $x = x^- \Leftrightarrow x = \mathbf{0}$. Hence an extremal 2-independent set in IF_p^n has $2 + (p^n - 2)/2$ members for even p and $1 + (p^n - 1)/2$ for odd p . An extremal 3-independent set in $\text{IF}_2^n = \{0, 1\}^n$ has 2^{n-1} members, an example of which is $\{1\} \times \{0, 1\}^{n-1}$.

Assume henceforth that $(k, p) = (3, 3)$. The j^{th} components of $x, y, z \in \text{IF}_3^n$ sum to 0 if and only if they are 0, 0, 0 or 1, 1, 1 or 2, 2, 2 or a permutation of 0, 1, 2. Hence $S \subseteq \text{IF}_3^n$ is 3-independent if and only if no three *distinct* members of S sum to $\mathbf{0}$. Since $\{0, 1\}^n$ is clearly 3-independent in $\{0, 1, 2\}^n$, $2 \leq c \leq 3$. Also note that if S_1 and S_2 are independent (we omit the “3-” subsets of IF_3^m and IF_3^n respectively, then $S_1 \times S_2$ is an independent subset of IF_3^{m+n} . Hence $f(m+n) \geq f(m)f(n)$, and it follows that for every $\epsilon > 0$ there is an n_ϵ such that $c_n > c - \epsilon$ for all $n > n_\epsilon$.

We say that $S \subseteq \text{IF}_3^n$ is *invertible* if $x \in S \Rightarrow x^- \in S$. Let $\langle x, y \rangle$ denote the inner product (mod 3) of $x, y \in \text{IF}_3^n$, and let T denote the family of all $t = (t_1, \dots, t_n)$ in $(\text{IF}_3^n)^n$ such that the transformation

$$t(x) = (\langle t_1, x \rangle, \dots, \langle t_n, x \rangle)$$

is a bijection on IF_3^n . Examples are $t_j = (0, \dots, 0, 1_j, 0, \dots, 0)$ for all j , whence $t(x) = x$, and $t = ((1, 0, \dots, 0), (1, 1, 0, \dots, 0), \dots, (1, 1, \dots, 1))$. We say that $S \subseteq \text{IF}_3^n$ is *closed under inversion* if there is a $t \in T$ such that either $t(S)$ is invertible or for each j there is a permutation $\sigma_j: \{0, 1, 2\} \rightarrow \{0, 1, 2\}$ such that the set obtained by applying σ_j to the j^{th} components of the members of $t(S)$ for $j = 1, \dots, n$ is invertible. Most of the large independent sets that we describe are closed under inversion, but some extremal independent sets are not. We note in the next section that no extremal set for $n = 3$ is closed under inversion.

2. Large Independent Sets for Small n

Let $* = \{1, 2\}$. A row with h $*$'s in an S display stands for 2^h members of S according to the $*$ realizations, unless noted otherwise. We refer to a line or row of a display matrix as a ROW.

This section notes that $f(2) = 4$, $f(3) = 9$, $f(4) \geq 20$, $f(5) \geq 42$, $f(6) \geq 112$ and $f(7) \geq 236$. Best prior lower bounds at $n = 5$ and $n = 7$ were 40 and 224 respectively. We also comment on closure under inversion.

Figure 1 displays large independent sets for $n \in \{2, 3, 4\}$. We often leave independence verification to inspection (no three rows sum to $\mathbf{0}$), but make a few observations. For example, a ROW with only 0's and $*$'s cannot violate independence. Independence is not violated in the first $n = 4$ display in Figure 1 with the choice of two top rows ($*$ in column 1) and one bottom row (0 in column 1) since the two top rows will have $\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}$ in a column whose entry for the bottom row is 1 or 2.

A subset S of IF_3^2 displayed as

00	01	02
10	11	12
20	21	22

is independent if and only if it does not include all three entries in some row, or in some column, or include three that cover the rows and columns. This leaves various patterns that verify $f(2) = 4$, all of which map into $**$ (top line, Figure 1) by transformations in T and within-column permutations. Hence all extremal (independent) sets for $n = 2$ are closed under inversion.

We can build on the $n = 2$ results to obtain $f(3) = 9$. The first matrix for $n = 3$ on Figure 1 puts together two extremal sets from $n = 2$ (columns 2 and 3) and adds a ninth row 212 when it is observed that no 2-row sum from the first eight rows is 121. We omit the details which show that all extremal sets for $n = 3$ are isomorphic under transformations in T plus within-column and among-column permutations. The transformation used on Figure 1 takes the first matrix into one that has the same frequency of 0's, 1's, and 2's in each column. No extremal set for $n = 3$ can be closed under inversion. Otherwise, because $f(3)$ is odd, an invertible image set must contain $\mathbf{0}$, which in conjunction with an inverse pair $\{x, x^-\}$ from the image set, $x \neq \mathbf{0}$, would give $\mathbf{0} + x + x^- = \mathbf{0}$.

Two rather different independent sets of order 20 for $n = 4$ complete Figure 1. The second, which was composed from two extremal sets for $n = 3$ (columns 2, 3, 4) is closed under inversion since its image at the right under the affine transformation $t(x) + (1, 1, 1, 1)$ is invertible.

The first matrix for $n = 4$ shows that no violation of independence for the S set of Figure 2 occurs within either the upper ($*$ in column 1) or lower (0 in column 1) part of its matrix. If we look for a violation in the two parts jointly, columns 1 and 2 show that we must, in effect, consider either A or B at the right of Figure 2. Since the final ROW of B can be used for only one of three distinct vectors that might sum to $\mathbf{0}$, it is tantamount to A , which is independent by the first case for $n = 4$ in Figure 1.

$$\begin{array}{c} n = 2 \\ f(2) = 4 \end{array}
 \begin{array}{c|c} 1 & 2 \\ \hline 0 & 0 \\ 1 & 1 \\ 2 & 1 \\ 0 & 2 \end{array}
 \xrightarrow[t_1 = 11, t_2 = 21]{}
 \begin{array}{c|c} 1 & 2 \\ \hline 0 & 0 \\ 2 & 0 \\ 0 & 2 \\ 2 & 2 \end{array}
 \xrightarrow[\text{interchange } 0 \& 1]{}
 \begin{array}{c|c} 1 & 2 \\ \hline 1 & 1 \\ 2 & 1 \\ 1 & 2 \\ 2 & 2 \end{array}
 = \begin{array}{c|c} & \\ \hline * & * \end{array}$$

$$\begin{array}{c} n = 3 \\ f(3) = 9 \end{array}
 \begin{array}{c|c|c} 1 & 2 & 3 \\ \hline 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ \hline 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \\ \hline 2 & 1 & 2 \end{array}
 \xrightarrow[t_1 = 100, t_2 = 010, t_3 = 101]{}
 \begin{array}{c|c|c} 1 & 2 & 3 \\ \hline 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ \hline 1 & 0 & 1 \\ 1 & 0 & 2 \\ 1 & 1 & 0 \\ 1 & 2 & 0 \\ \hline 2 & 1 & 1 \end{array}$$

$$\begin{array}{c} n = 4 \\ |S| = 20 \end{array}
 \begin{array}{c|c|c|c} 1 & 2 & 3 & 4 \\ \hline * & 0 & 0 & 0 \\ * & 0 & * & 0 \\ 0 & 0 & 0 & * \\ \hline 0 & * & * & * \end{array}
 \xrightarrow[t_1 = 1000, t_2 = 1100, t_3 = 2110, t_4 = 0111, \text{ AND add 1 to all entries to get invertible}]{}
 \begin{array}{c|c|c|c} 1 & 2 & 3 & 4 \\ \hline 0 & 0 & 0 & 0 \\ & 0 & 1 & 0 \\ & 1 & 1 & 1 \\ & 0 & 2 & 2 \\ & 1 & 2 & 0 \\ & 2 & 2 & 2 \\ & 2 & 2 & 2 \\ \hline 1 & 1 & 1 & 1 \\ & 1 & 2 & 1 \\ & 0 & 2 & 1 \\ & 2 & 2 & 2 \\ & 0 & 1 & 0 \\ & 0 & 2 & 0 \\ & 1 & 0 & 0 \\ & 1 & 1 & 1 \\ & 2 & 1 & 1 \\ \hline 2 & 1 & 1 & 1 \\ & 2 & 1 & 1 \end{array}
 \xrightarrow{}
 \begin{array}{c|c|c|c} 1 & 2 & 3 & 4 \\ \hline 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 \\ 1 & 2 & 2 & 1 \\ 1 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 \\ \hline 2 & 2 & 1 & 0 \\ 2 & 2 & 1 & 1 \\ 2 & 2 & 2 & 1 \\ 2 & 2 & 2 & 2 \\ 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 \\ 2 & 1 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ \hline 0 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \end{array}$$

Figure 1. Large independent sets for $n \leq 4$.

The four top matrices of Figure 3 display independent sets in IF_3^6 . All are closed under inversion. The 112-element sets α and β are essentially due to Hill [6] (see also [11, p. 113]) and are similar up to permutations of rows and columns. Since Hill proved that a

1	2	3	4	5	Submatrices															
*			*	0	0	1	1		*	0	0	*	1		*	0	0			
		1	0	*	0				0	*	0				0	*	0	0	*	0
			0	0	*				0	0	*				0	0	*		0	0
	2	*	*	*		2	2	*	*	*										
0			*	0	0			*	0	0										
		0	0	*	0			0	*	0										
			0	0	*			0	0	*										
		1	*	*	*			0	0	*										
S = 42					A					B										

$|S| = 42$

A

B

Figure 2. $f(5) \geq 42$.

cap in $\text{PG}(5, 3)$ contains at most 56 points, they are extremal when S is restricted to be closed under inversion, but it is not known if they are extremal otherwise. The bottom matrix on Figure 3 places copies of α , β and γ in columns 2 through 7 for $n = 7$ and prefaces them by 0, 1 and 2 respectively in column 1. The results for $n = 6$ show that one vector from each of the three ROWs of the bottom matrix, say x for α , y for β , and z for γ , is needed to refute independence via $x + y + z = \mathbf{0}$. Let $C = \{0, 1, 2\}$. Then with $x \in \alpha$ and $y \in \beta$, the possibilities for $x + y$ lie in (up to permutations of the last six columns)

- * * * * *
- or C * * * * 0
- or C C * * 0 0
- or C C C * * *
- or C C C C C C (x and y from δ only)

and in the last case ($CC \cdots C$) the number of 0's must be even. Hence, with $z \in \gamma$, we cannot have $x + y + z = \mathbf{0}$ when $x + y \in CC \cdots C$. Moreover, z added to any $x + y$ for the first four cases cannot give $\mathbf{0}$. Hence the matrix satisfies independence, so $f(7) \geq 236$.

In the preceding analysis, all largest independent sets for even n are closed under inversion and, as far as we can tell, none for odd n is closed under inversion. To four decimal places, the corresponding values for the c_n are

$$\begin{array}{ll}
 c_2 = 2.0000 & c_5 \geq 2.1118 \\
 c_3 = 2.0801 & c_6 \geq 2.1955 \\
 c_4 = 2.1147 & c_7 \geq 2.1827.
 \end{array}$$

We now give constructions which show that $c > c_6$.

$$\gamma = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline * & 0 & 0 & 0 & 0 & 0 \\ 0 & * & 0 & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & 0 & * & 0 & 0 \\ 0 & 0 & 0 & 0 & * & 0 \\ 0 & 0 & 0 & 0 & 0 & * \end{array}$$

$|\gamma| = 12$

$$\delta = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline * & * & * & * & * & * \end{array}$$

subject to
an even number of 1's in each row

$|\delta| = 32$

$$\alpha = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline & 0 & * & * & * & 0 \\ & 0 & 0 & * & * & * \\ 0 & * & 0 & 0 & * & * \\ & * & * & 0 & 0 & * \\ & * & * & * & 0 & 0 \\ \hline & * & 0 & * & 0 & 0 \\ & * & 0 & 0 & * & 0 \\ * & 0 & * & 0 & * & 0 \\ & 0 & * & 0 & 0 & * \\ & 0 & 0 & * & 0 & * \\ \hline & \delta \end{array}$$

$|\alpha| = 112$

interchange
0 & * above
 δ

$\longrightarrow$

$$\beta = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline & * & 0 & 0 & 0 & * \\ & * & * & 0 & 0 & 0 \\ * & 0 & * & * & 0 & 0 \\ & 0 & 0 & * & * & 0 \\ & 0 & 0 & 0 & * & * \\ \hline & 0 & * & 0 & * & * \\ & 0 & * & * & 0 & * \\ 0 & * & 0 & * & 0 & * \\ & * & 0 & * & * & 0 \\ & * & * & 0 & * & 0 \\ \hline & \delta \end{array}$$

$|\beta| = 112$

$f(6) \geq 112$

$f(7) \geq 236 :$

1	2-7	
0	α	112
1	β	112
2	γ	12
		236

Figure 3. $f(6) \geq 112, f(7) \geq 236 : \alpha \cap \beta = \delta, \gamma \cap (\alpha \cup \beta) = \emptyset$.

3. New Lower Bounds

The Cartesian product of α with itself K times gives $f(6K) \geq 112^K$, hence $c_{6K} \geq 112^{1/6} \doteq 2.1955$. We can do better for some K by replacing one α in the product by γ and taking the union of this with similar products that involve β . Let

$$M_K = \begin{array}{c} \begin{array}{cccccc} & \text{6K columns} \\ \gamma & \alpha & \alpha & \alpha & \dots & \alpha \\ \beta & \gamma & \alpha & \alpha & \dots & \alpha \\ \beta & \beta & \gamma & \alpha & \dots & \alpha \\ \vdots & & & & & \\ \beta & \beta & \beta & \beta & \dots & \beta & \gamma \\ \delta & \delta & \delta & \delta & \dots & \delta & \delta \end{array} \end{array}, \quad K \geq 2.$$

The addition of δ^K does not violate independence and increases the total number of rows, but only slightly:

$$\begin{aligned} |M_K| &= K(12)(112)^{K-1} + 32^K \\ &= 32^K[(3K/8)(7/2)^{K-1} + 1]. \end{aligned}$$

We claim that M_K is an independent set in IF_3^{6K} . Let $x \leq y \leq z$ ($\leq$ is *not below*) be rows of M_K . If x, y and z are in the same ROW and $x + y + z = \mathbf{0}$, then $x = y = z$ within each α, β, γ or δ , so $x = y = z$ overall. Assume $x < y < z$ henceforth. Suppose two are in one ROW and the third is in another ROW. Then in some 6-column block we see

$$\begin{array}{ccc} x, y: & \gamma & \gamma \\ & \text{or} & \\ z: & \beta & \delta \end{array} \quad \text{OR} \quad \begin{array}{ccc} x: & \alpha & \gamma \\ & \text{or} & \\ y, z: & \gamma & \delta \end{array}.$$

Since every ROW in α, β and δ has at least three *'s, and the sum of two in γ has at least four 0's, the first three cases have a column with two 0's and a *, so $x + y + z \neq \mathbf{0}$. The case of (γ, δ, δ) is covered by the $CC \cdots C$ analysis at $n = 7$. Suppose x, y and z are in different ROWS of M_K . Then some 6-column block has (α, γ, β) or (α, γ, δ) for (x, y, z) and, since $\delta \subset \beta$, we get $x + y + z \neq \mathbf{0}$ by the $n = 7$ analysis of the preceding section.

Since M_K is an independent set, we have

$$c_{6K} \geq 32^{1/6}[(3K/8)(7/2)^{K-1} + 1]^{1/6K}.$$

Values of this lower bound for selected K are

2.1940369 for $K = 9$

2.1980407 for $K = 10$

2.2099617 for $K = 24$

2.2099834 for $K = 25$

2.2099807 for $K = 26$.

So this scheme first passes $112^{1/6}$ at $K = 10$ ($n = 60$) and realizes its maximum of nearly 2.21 at $K = 25$ ($n = 150$).

We note for the record that if distinct x , y and z have a 6-column block with pattern

$$\begin{array}{cccccc} \alpha & & \alpha & & \beta & & \alpha & & \beta & & \delta & & \delta \\ \beta & \text{or} & \gamma & \text{or} & \gamma & \text{or} & \gamma & \text{or} & \gamma & \text{or} & \gamma & \text{or} & \delta \\ \gamma & & \delta & & \delta & & \gamma & & \gamma & & \gamma & & \gamma \end{array} \quad (1)$$

or a permutation thereof, then $x + y + z \neq \mathbf{0}$.

To improve the bound of 2.2099 ... on c , we take our scheme up a level. Let

$$\begin{array}{l} \Gamma_K = \begin{array}{c} \text{6K columns} \\ \begin{array}{|cccc|} \hline \gamma & \gamma & \alpha & M_{K-3} \\ \gamma & \beta & \gamma & M_{K-3} \\ \alpha & \gamma & \gamma & M_{K-3} \\ \hline \end{array} \end{array} \\ \Delta_K = \begin{array}{c} \text{6K columns} \\ \begin{array}{|cc|} \hline \delta & \delta & M_{K-2} \\ \hline \end{array} \end{array} \end{array} \quad M'_K = \begin{array}{c} \text{6K columns} \\ \begin{array}{|cccccc|} \hline \gamma & \beta & \beta & \beta & \dots & \beta \\ \alpha & \gamma & \beta & \beta & \dots & \beta \\ \alpha & \alpha & \gamma & \beta & \dots & \beta \\ \vdots & & & & & \\ \vdots & & & & & \\ \alpha & \alpha & \alpha & \alpha & \dots & \alpha & \gamma \\ \delta & \delta & \delta & \delta & \dots & \delta & \delta \\ \hline \end{array} \end{array}$$

and note that each of Γ_K (see preceding paragraph), Δ_K and M'_K (see M_K) are independent, with $\Gamma_K \cap \Delta_K = \emptyset$. Let

$$M_{K,J} = \begin{array}{c} J(6K) \text{ columns} \\ \begin{array}{|cccccc|} \hline \Gamma_K & M_K & M_K & M_K & \dots & M_K \\ M'_K & \Gamma_K & M_K & M_K & \dots & M_K \\ M'_K & M'_K & \Gamma_K & M_K & \dots & M_K \\ \vdots & & & & & \\ M'_K & M'_K & M'_K & M'_K & \dots & \Gamma_K & M_K \\ \Delta_K & \Delta_K & \Delta_K & \Delta_K & \dots & \Delta_K & \Delta_K \\ \hline \end{array} \end{array}$$

with

$$\begin{aligned}
 |M_{K,J}| &= J(3)(12^2)(112) |M_{K-3}| |M_K|^{J-1} + [32^2 |M_{K-2}|]^J \\
 &= (32)^{KJ} \left\{ \left[\left(\frac{189J}{128} \right) \left[\left(\frac{3K}{8} \right) \left(\frac{7}{2} \right)^{K-1} + 1 \right]^{J-1} \left[\left(\frac{3(K-3)}{8} \right) \left(\frac{7}{2} \right)^{K-4} + 1 \right] \right. \right. \\
 &\quad \left. \left. + \left[\left(\frac{3(K-2)}{8} \right) \left(\frac{7}{2} \right)^{K-3} + 1 \right]^J \right\}.
 \end{aligned}$$

We claim that $M_{K,J}$ is an independent set in $\mathbb{F}_3^{6JK}$. If x, y and z are in the same ROW of $M_{K,J}$ and $x + y + z = \mathbf{0}$ then $x = y = z$. Assume henceforth that different ROWS are used. If Δ'_K is involved, use a column block at the Γ_K position for another involved row. If Δ'_K is not involved, and x, y and z are in only two ROWS, look at the column block that involves Γ_K twice. If Δ'_K is not involved and x, y and z are in three different ROWS, use the column block with Γ_K in the middle of M_K and M'_K (vertically). In each case there will be a triplet pattern (vertically) in the first, second or third 6-column block of the focal Γ_K super-column block which adheres to (1), so in all cases $x + y + z \neq \mathbf{0}$. We omit the straightforward details.

Computations show that $|M_{K,J}|^{1/6JK}$ is maximized in the vicinity of $(K, J) = (25, 90)$, i.e., $n = 13500$, where it is slightly larger than 2.210147. Thus

$$c > 2.210147.$$

References

1. R.C. Bose, Mathematical Theory of the Symmetric Factorial Design, *Sankhya*, Vol. 8 (1947), pp. 107-166.
2. B. Segré, Le Geometrie di Galois, *Ann. Mat. Pura. Appl.*, Vol. 48 (1959), pp. 1-97.
3. B. Segré, Introduction to Galois Geometries, *Atti. Acad. Naz. Lincei Mem.*, Serie VIII, Vol. VIII (1967).
4. B. Segré, Forme e Geometrie Hermitiane, con Particolare Riguardo al Caso Finito, *Ann. Mat. Pura. Appl.*, Vol. 70 (1965), pp. 1-202.
5. J. McLaughlin, A Simple Group of Order 898, 128, 000, *Theory of Finite Groups*, Benjamin, New York, (1969), pp. 109-111.
6. R. Hill, On the Largest Size Cap in $S_{5,3}$, *Rend. Accad. Naz. Lincei*, Vol. 8, 54 (1973), pp. 378-384.
7. R. Hill, Caps and Groups, *Atti dei Convegni Lincei, Colloquio Internazionale sulle Teorie Combinatorie* (Roma 1973), No. 17 (Accad. Naz. Lincei, 1976), pp. 384-394.
8. R. Hill, Caps and Codes, *Discrete Math.*, Vol. 22 (1978), pp. 111-137.
9. A.A. Bruen and J.W.P. Hirschfeld, Applications of Line Geometry Over Finite Fields II. The Hermitian Surface, *Geom. Dedicata*, Vol. 7 (1978), pp. 333-353.
10. R.A. Games, The Packing Problem for Finite Projective Geometries, Thesis, The Ohio State University, 1980.
11. A.R. Calderbank and W.M. Kantor, The Geometry of Two-Weight Codes, *Bull. London Math. Soc.*, Vol. 18 (1986), pp. 97-122.
12. P. Frankl, R.L. Graham and V. Rödl, On Subsets of Abelian Groups With No 3-Term Arithmetic Progression, *J. Combin. Theory Ser. A*, Vol. 45 (1987), pp. 157-161.